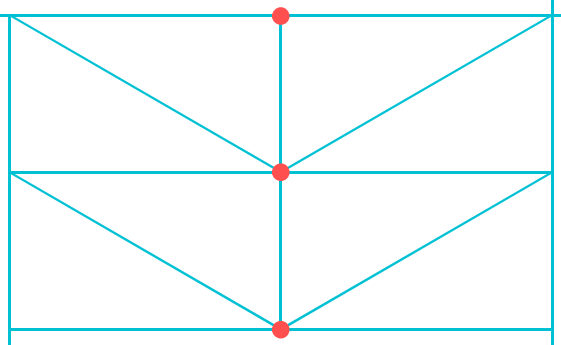
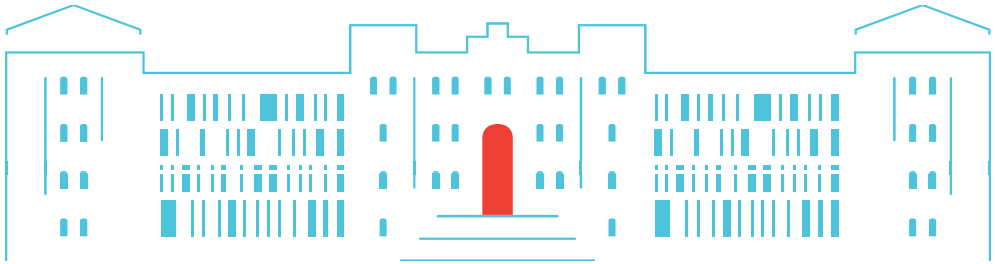
Distributed Systems
Lab 4
Vector Clocks

TUHH
Technische
Universität
Hamburg



3.12.25 and 5.12.25



Marcus Venzke

Agenda

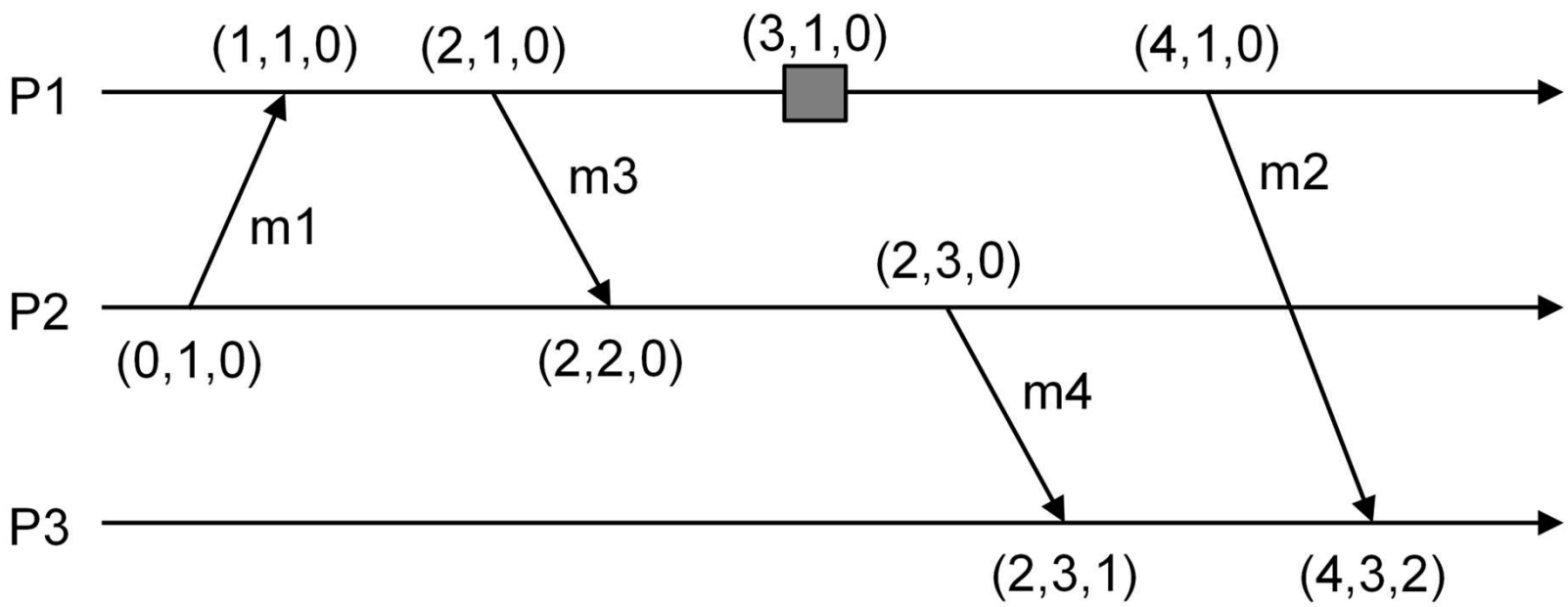
- Vector Clocks
- Causalities with Vector Clocks
- Tasks of Lab 4

TUHH

Vector Clocks

- Logical Clocks: No real time. Just causal event ordering.
- Vector Clock: Current time is a vector. One value per server.
- Event: Something happens on a server
  - In vector: Value of server increased by 1
- Message Sent
  - $\Rightarrow$ Event
  - Vector included in message
- Message Received
  - $\Rightarrow$ Event
  - Server's vector = Max (Server's vector, Vector in message)
  - E.g. Server P2, server's vector = [0, 1, 0]. Vector in message = [2,1,0]
    - Event: Increase server's vector to [0, 2, 0]
    - Max leads to [2, 2, 0]

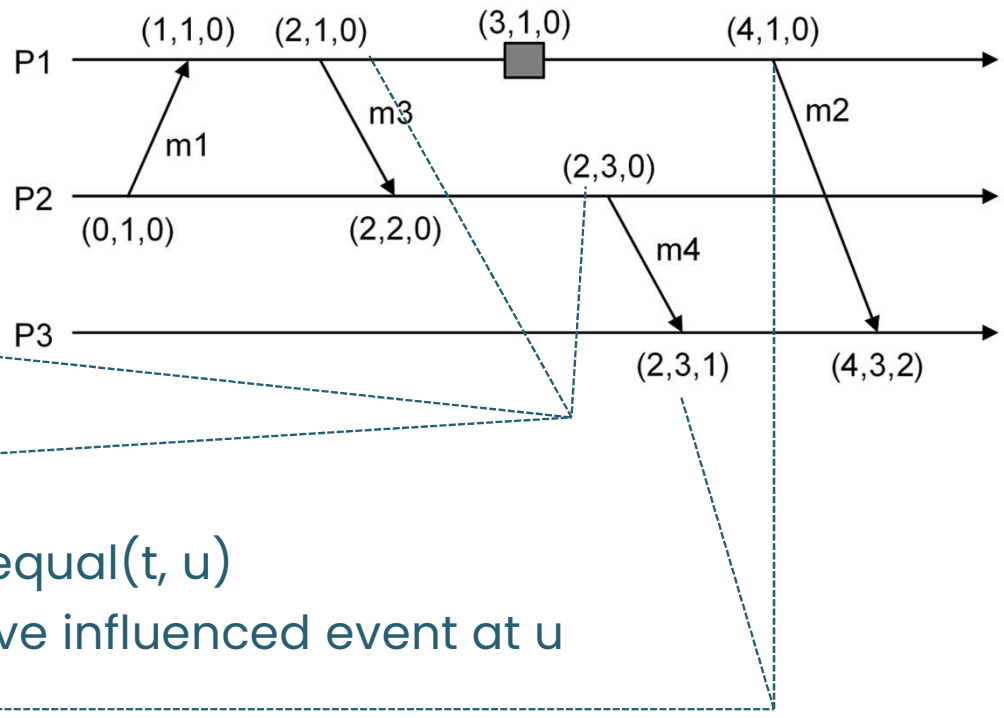
Vector Clocks – Example



From Figure 5.12 of book *Maarten van Steen, Andrew S. Tanenbaum: Distributed Systems, Fourth edition (4.03), self-published, January 2025.*

Causalities with Vector Clocks

- equal(t, u)
  - $\forall\, i: t_i = u_i$
  - Vector is exactly the same
  - Same point in time
- smallerOrEqual(t, u)
  - $\forall\, i: t_i \leq u_i$
- smaller(t, u)
  - smallerOrEqual(t, u) and not equal(t, u)
  - Causality: Event at t might have influenced event at u
- concurrent(t, u)
  - not smallerOrEqual(t, u) and not smallerOrEqual(u, t)
  - Causality: Event at t cannot have influenced event at u and vice versa



Task 1 – Implement Vector Clock as Class

- Implement `VectorClock.py`
  - Class clock with given methods
    - `eventHappens()` – Increase server's element in vector clock
    - `getTimeNoEvent()` – Return vector clock as list
    - `getTime()` – Do event + return vector clock as list
    - `updateTime(timeFromOtherServer)`
      - Do event + update times from other servers (max value)
  - Functions to compare times. Parameters = lists
    - `equal(time1, time2)` – Returns True only if two times are equal
    - `smallerEqual(time1, time2)`
      – Returns True only if time1 causally smaller or equal than time 1
    - `concurrent(time1, time2)`
      – Returns True only if time1 is causally concurrent to time
    - `totalOrder(time1, time2)` – Defines total order of times.
      - Returns -1 if time1 < time2, 0 if time1 == time2, 1 if time1 > time2
- Use `VectorClockTest.py` to test class

Task 2 – Integrate Time to Proxy and Stub

- Times added to each message between servers
  - E.g. Add "TIME":[10, 14, 4, 6] to dictionary of message (JSON)
  - `getTime()` for sending message, `updateTime()` for received message
  - In `AsyncBoardProxy.py` + stub (`BoardServer.py`)

- Issue: Responses are not JSON
  - Change protocol, so response are JSON messages
  - Issue2: Client proxy (`BoardProxy.py`) does not understand JSON responses
    - Also update `BoardProxy.py` for JSON response
    - Or: Implement stub to not send old responses to clients

recommended way to add the time to the response

Task 3 – Integrate Vector Clock to Message Board System

- Core Algorithm: Each server forwards updates to all servers
  - But: Each server sorts messages by time stamps
- a) Server receives put() from client
  - Time stamp added (e.g. [[9, 2, 2, 2], "Hi"])
  - Message + time stamp stored + forwarded to other servers
  - modify(): Same time stamp. Only message changed.
  - Do clients produce errors for messages such as [[9, 2, 2, 2], "Hi"]?
    - Correct it in clients
- b) Implement sorting in AsyncBoardStorage.py
  - Implement function defining order in InformAllOtherServersWithClock.py
  - Following order:
    - Ensure causal order (with functions from VectorClock.py)
    - Same order on all servers

Exercise Sheet

- Tasks described in detail on exercise sheet 4
- Download exercise sheet from StudIP

- You may start implementing Lab 4 after next slide
  - You can also solution of Lab 3
  - I will also answer questions to Lab 4

Next Week / In Two Weeks

- Next week: Time for questions
  - You may already submit your solutions

- In 2 weeks:
  - Last option to submit your solution of lab 3
  - New exercise sheet 4 starts
    - Implement logical clock (vector clock)
    - Use vector clock for message board

The End